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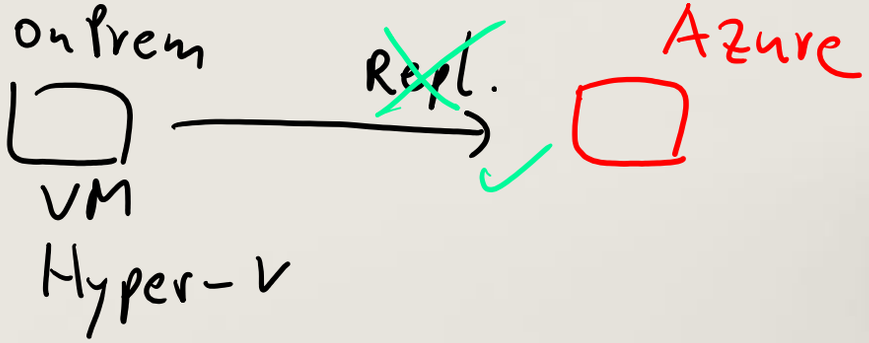
Lift & Shift



AZ-104

Administer Azure Virtual Machines

Azure Site Recovery (ASR)

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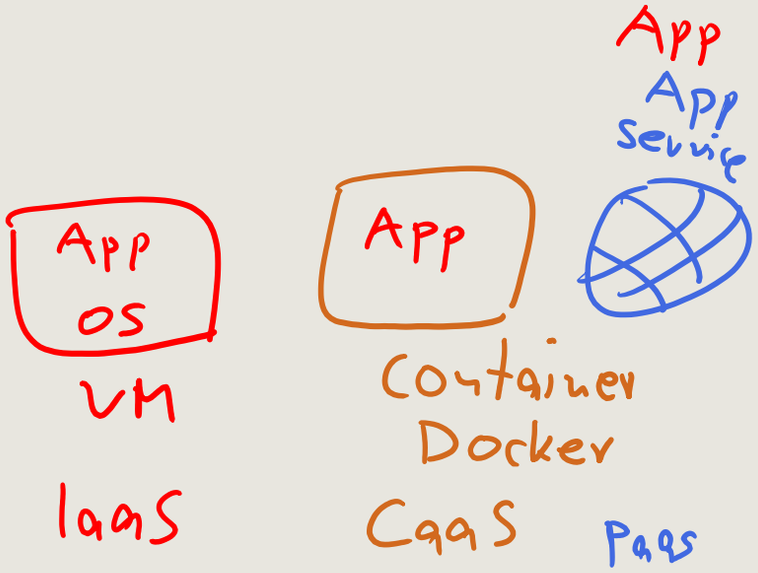
AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

IaaS



Learning Objectives - Administer Azure Virtual Machines

- Introduction to Azure Virtual Machines
- Configure Virtual Machine Availability Sets
- Lab 08 – Manage Virtual Machines

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VMware on Azure

Introduction to Azure Virtual Machines

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Learning Objectives – Introduction to Azure Virtual Machines

- Review Cloud Services Responsibilities
- Plan Virtual Machines
- Determine Virtual Machine Sizing
- Determine Virtual Machine Storage
- Connect to Virtual Machines
- Demonstration - Create and connect to virtual machines
- Learning Recap

Exam objectives

Create and configure virtual machines (VMs)

- Create a VM
- Move VMs between resource groups
- Manage VM sizes
- Manage VM disks

Configure secure access to virtual networks

- Implement Azure Bastion

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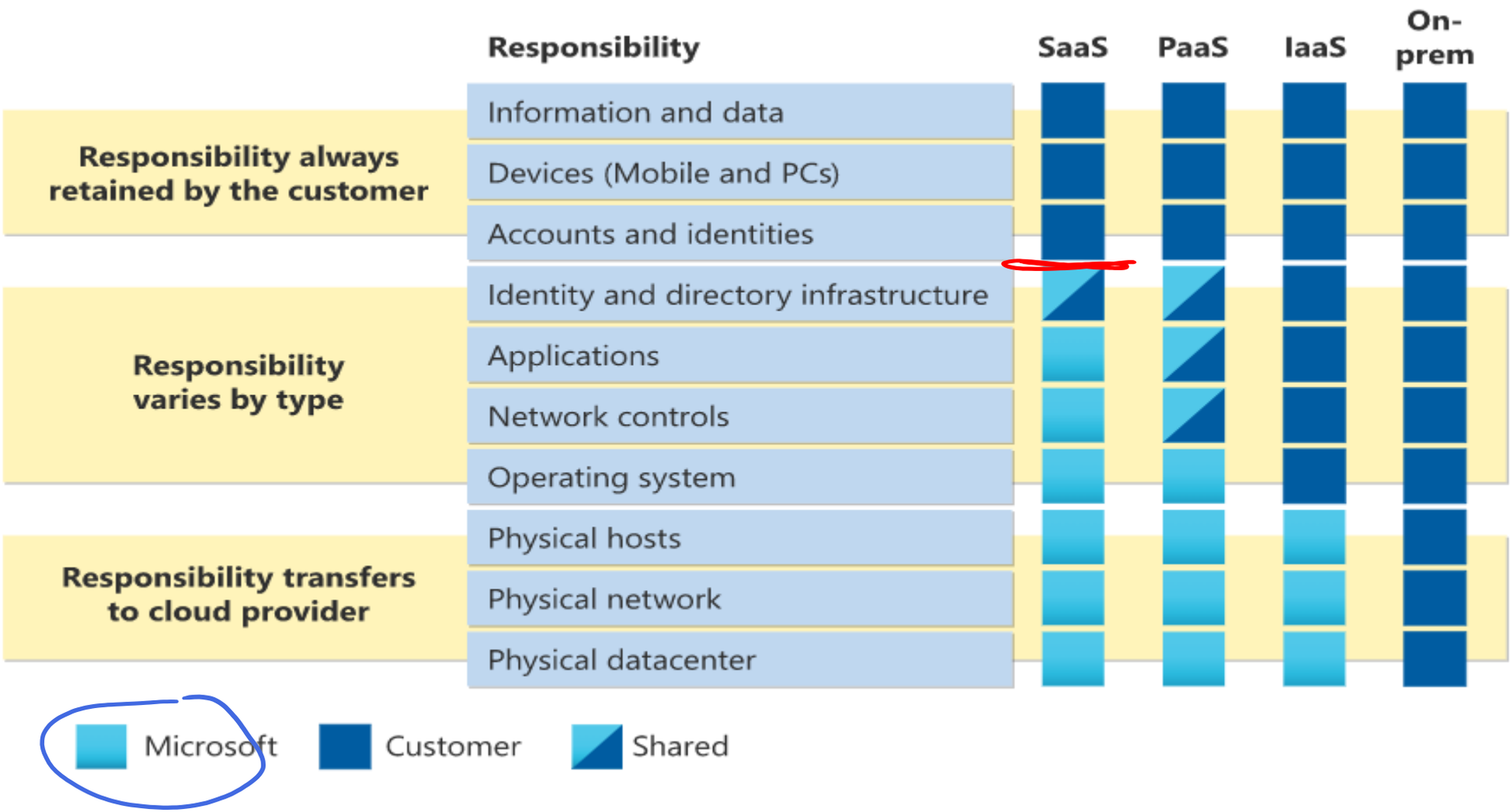
Review Cloud Services Responsibilities



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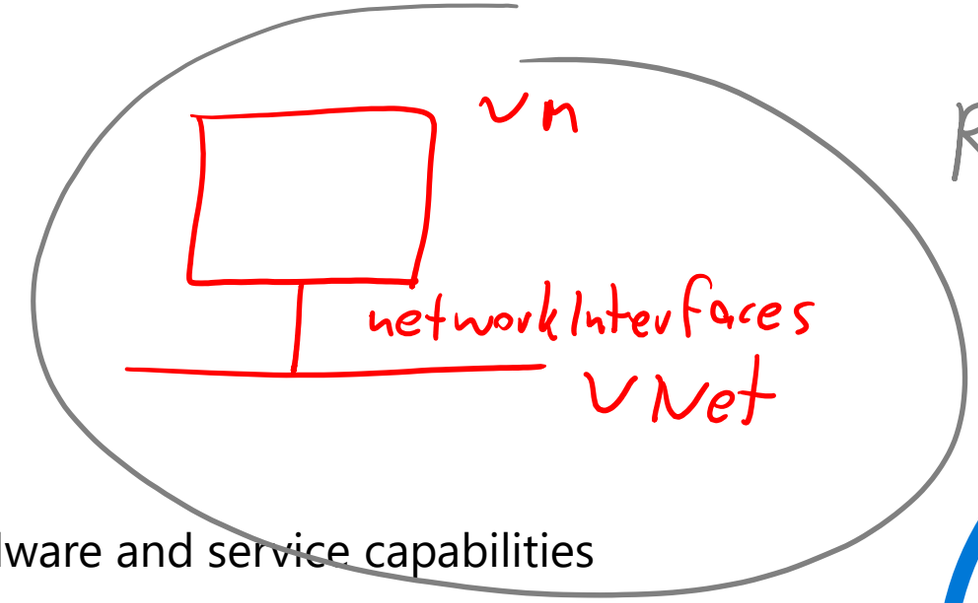
Plan Virtual Machines



Region

- Start with the network
- Name the virtual machine
- Choose a location
  - Each region has different hardware and service capabilities
  - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing

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Determine Virtual Machine Sizing

Type	Description
General purpose	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

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Determine Virtual Machine Storage

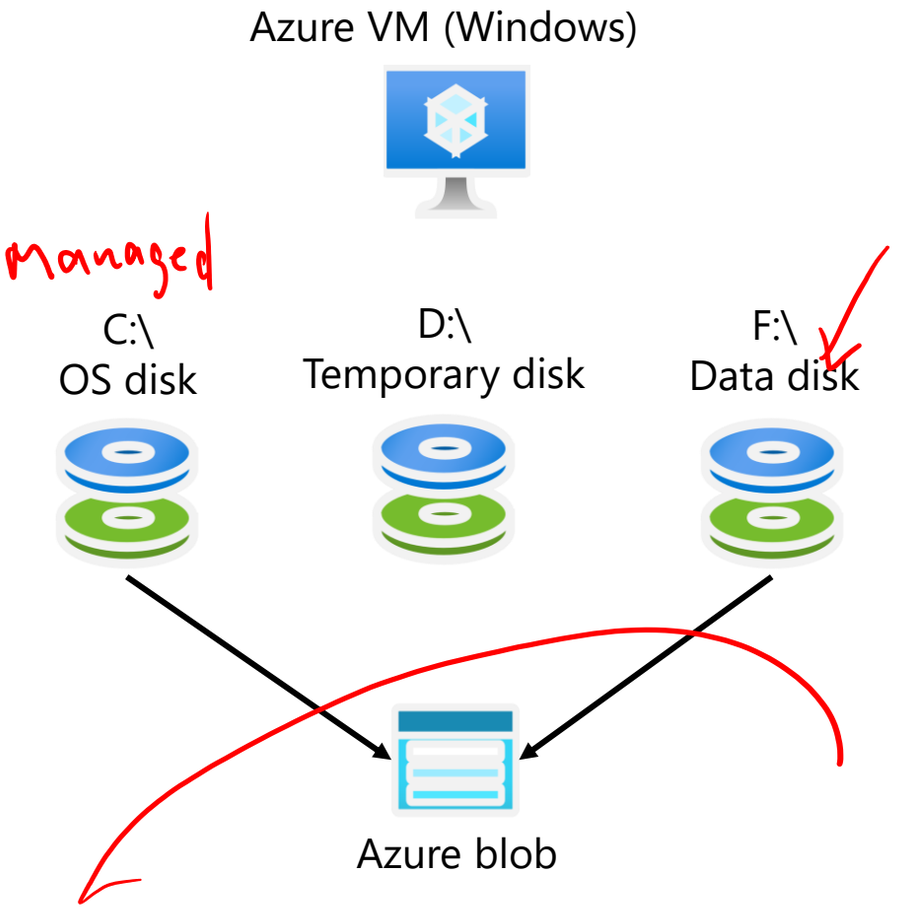
Each Azure VM has two or more disks:

- OS disk
- Temporary disk (not all SKUs have one, content can be lost)
- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks



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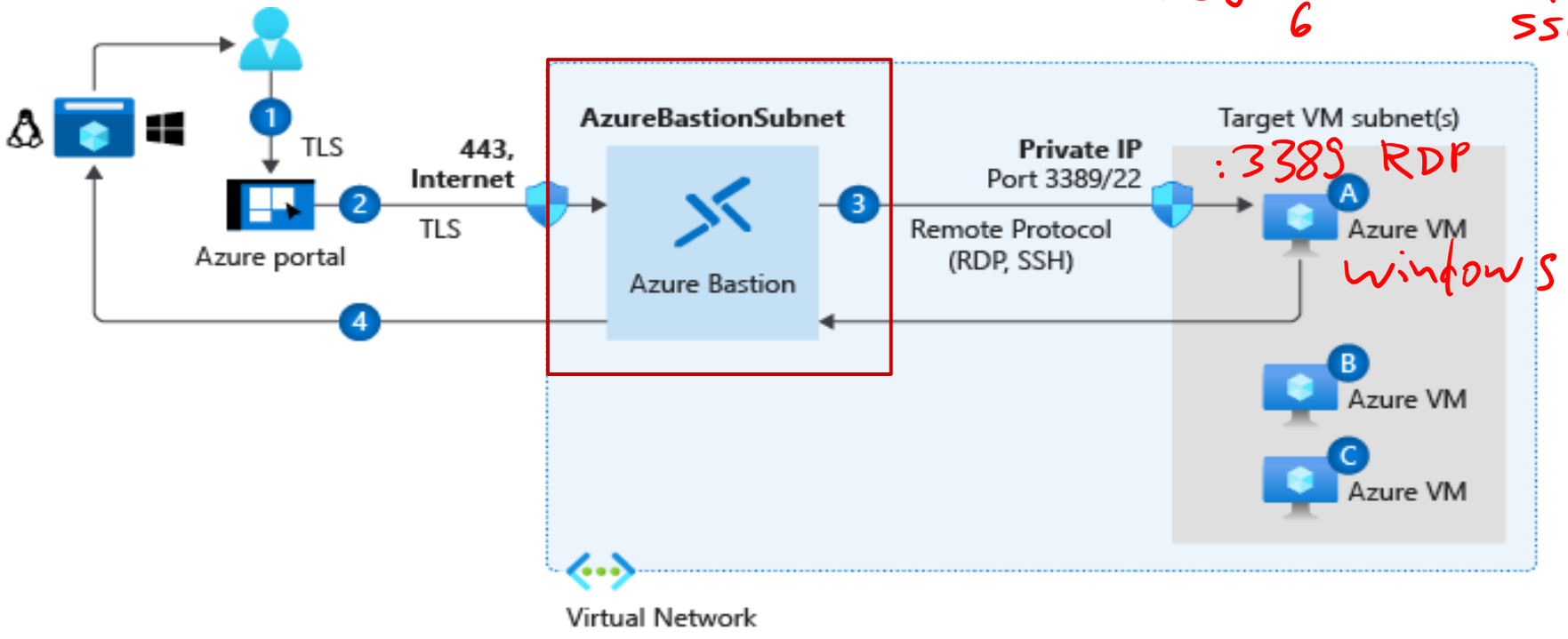
Connect to Virtual Machines

:5985 WinRM (remote PowerShell)
6 SSL



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

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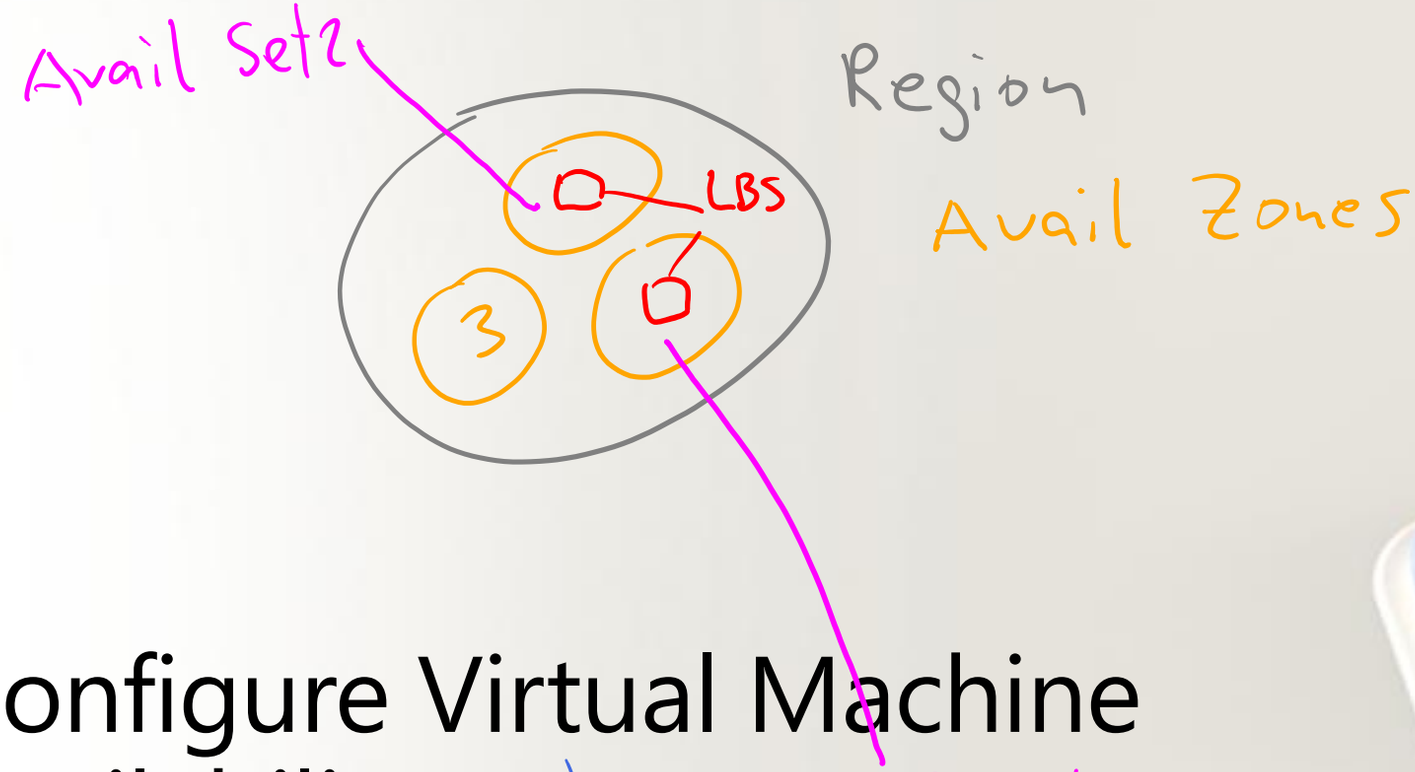

Configure Virtual Machine Availability

	Update Domain 1	Update Domain 2
Fault Domain 1	VM1	
FD 2		VM2

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Configure Azure Virtual Machine Availability Introduction

- Plan for Maintenance and Downtime
- Setup Availability Sets
- Review Update and Fault Domains
- Review Availability Zones
- Compare Vertical to Horizontal Scaling
- Create and Configure Scaling
- Demonstration – Virtual Machine Scaling
- Bring Azure innovation to your hybrid environments with Azure Arc (optional)
- Learning Recap

Exam objectives

Create and configure virtual machines

- Deploy virtual machines to availability zones and availability sets
- Deploy and configure an Azure Virtual Machine Scale Sets

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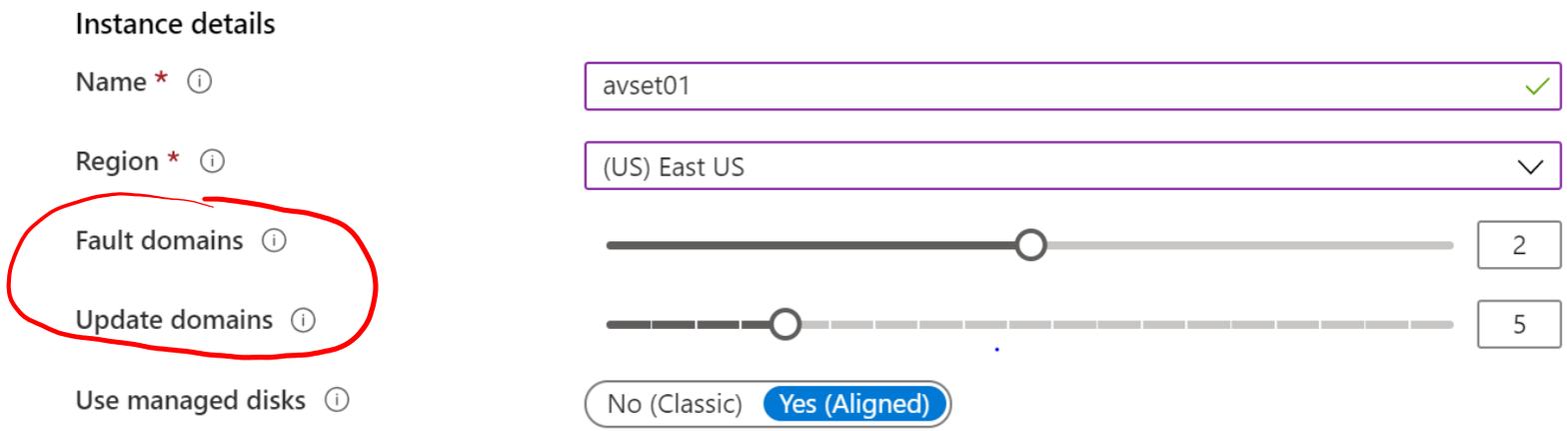
Setup Availability Sets



Two or more instances in Availability Sets = 99.95% SLA

Configure multiple Virtual Machines in an Availability Set

Configure each application tier into separate Availability Sets

Combine a Load Balancer with Availability Sets

Use managed disks with the Virtual Machines

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Review Availability Zones

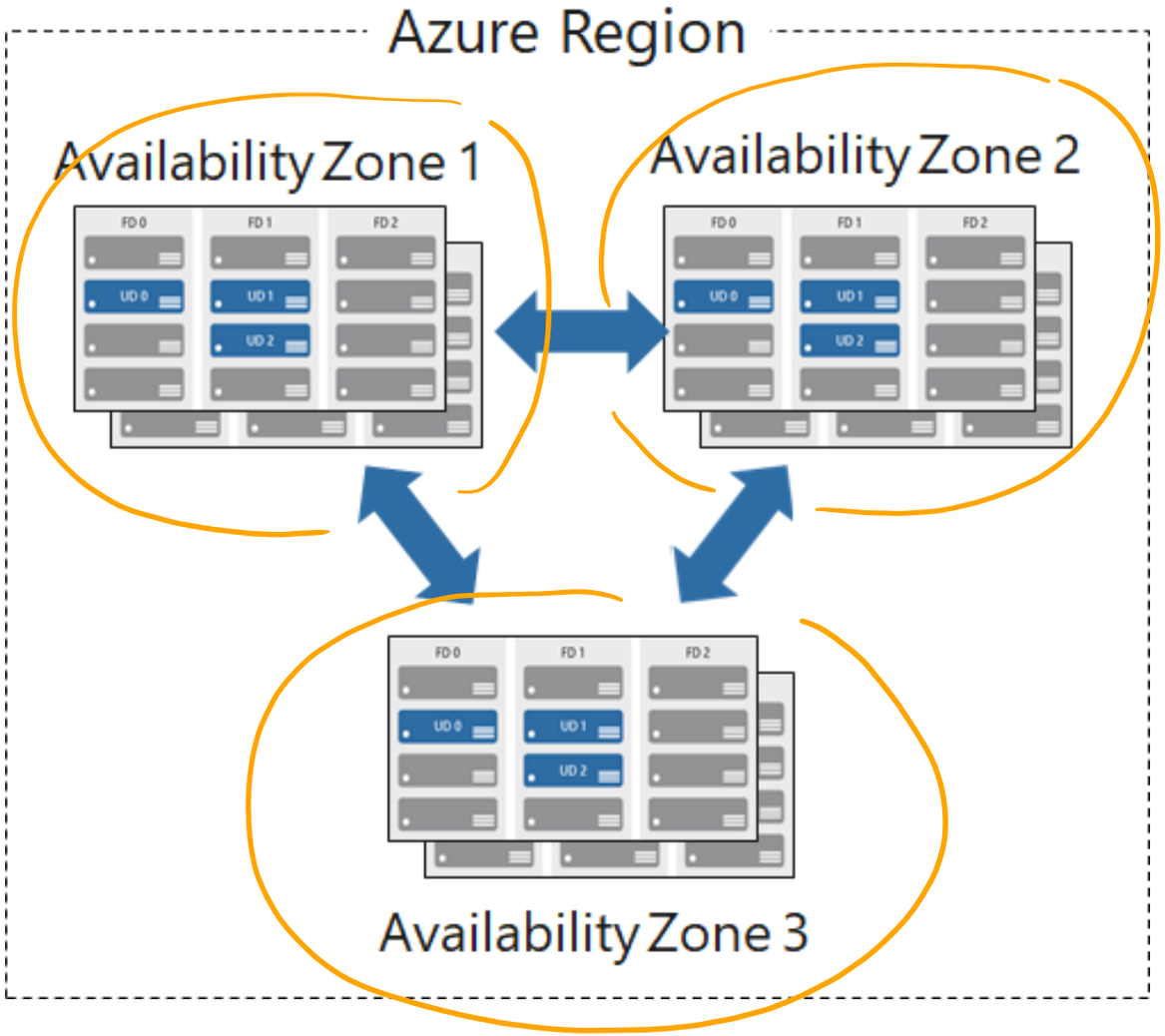
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

Combines update and fault domains

Provides 99.99% SLA



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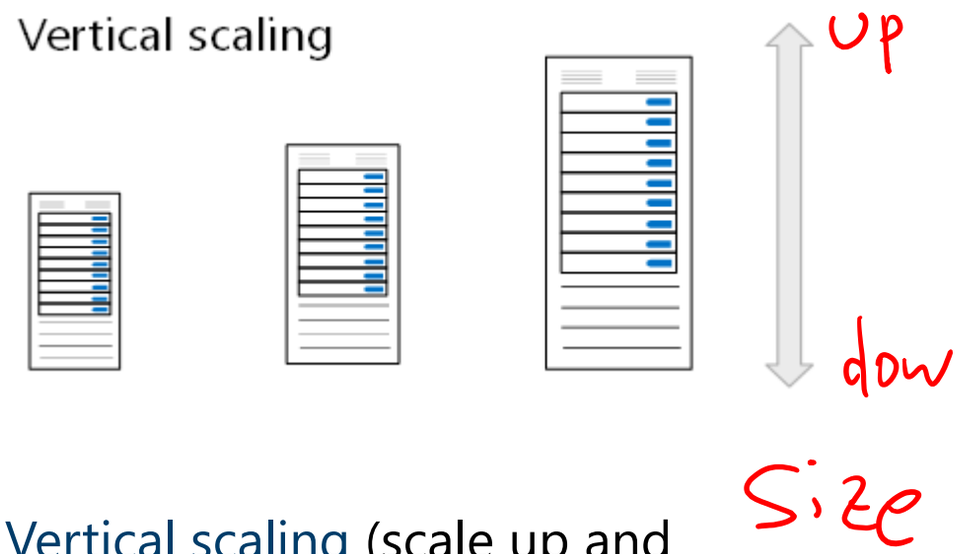
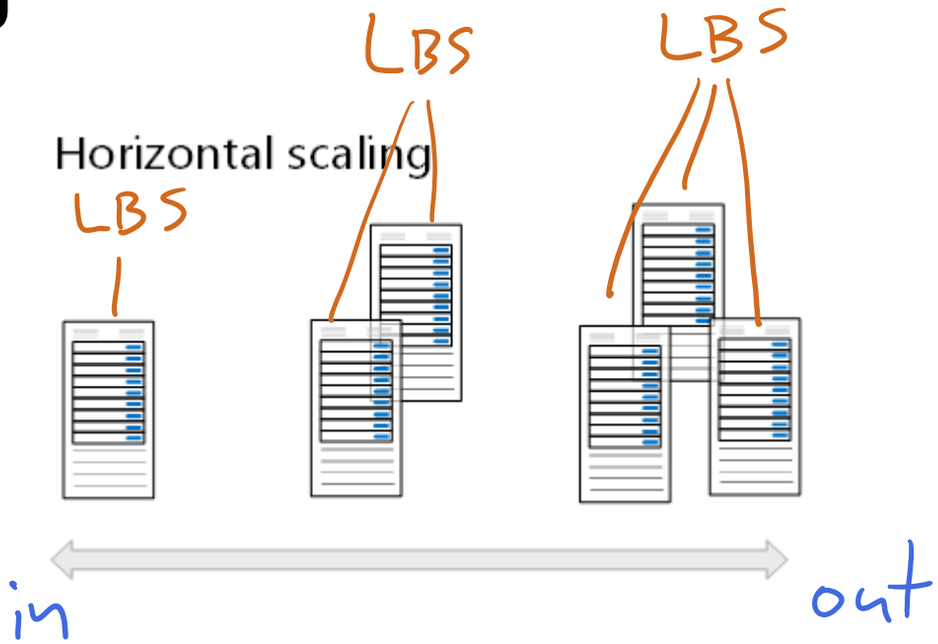
Compare Vertical to Horizontal Scaling





Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

down down

Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

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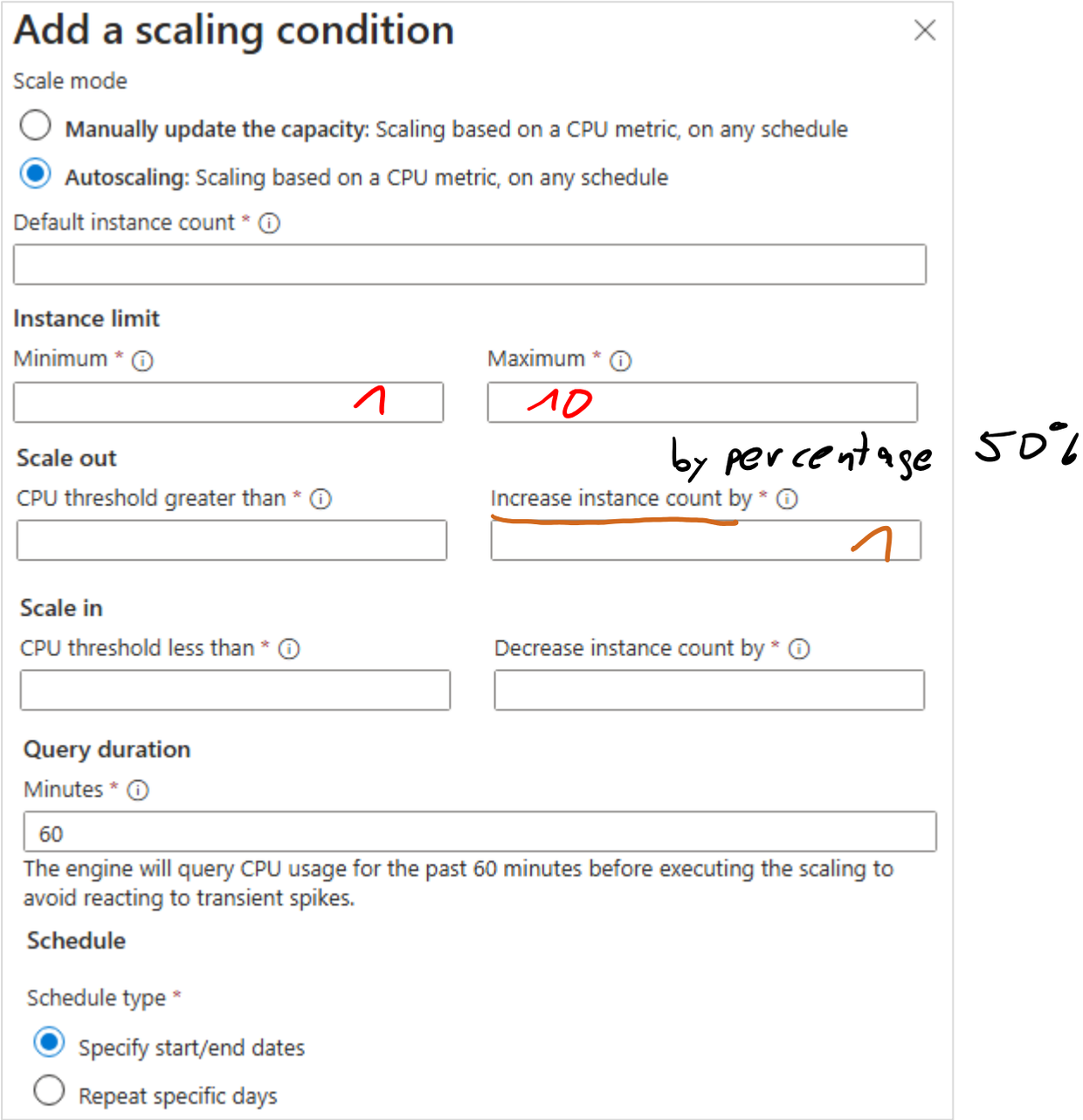
Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters



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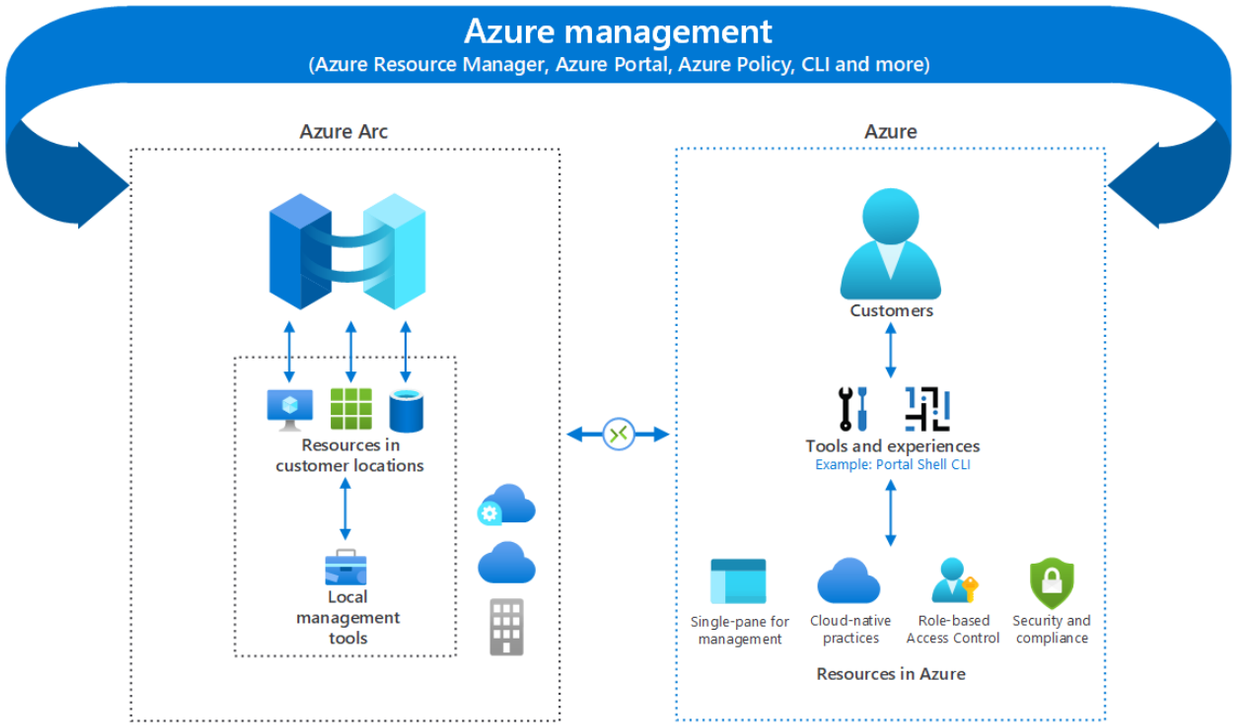
Bring Azure innovation to your hybrid environments with Azure Arc

- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment



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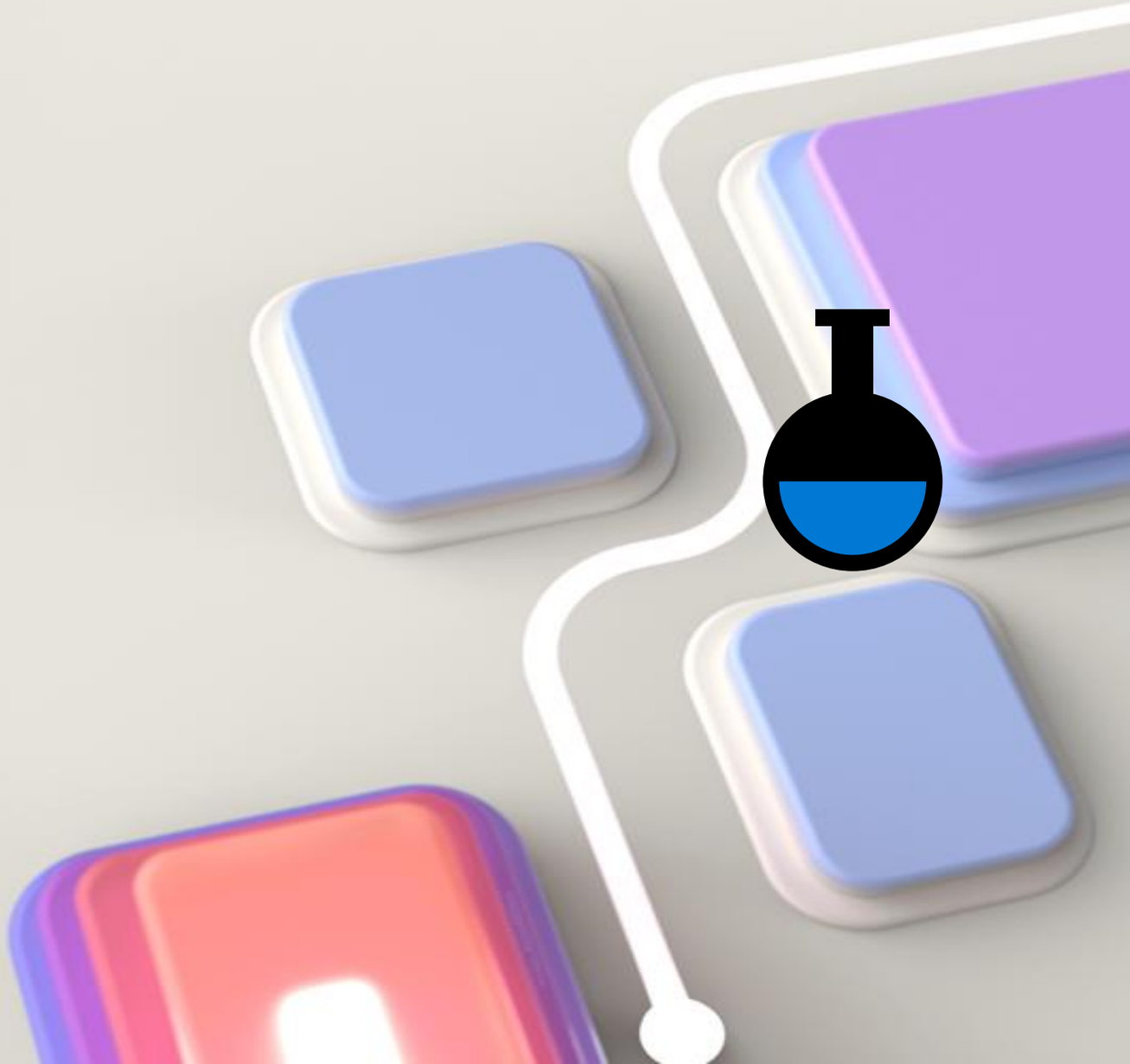
Lab – Manage Virtual Machines



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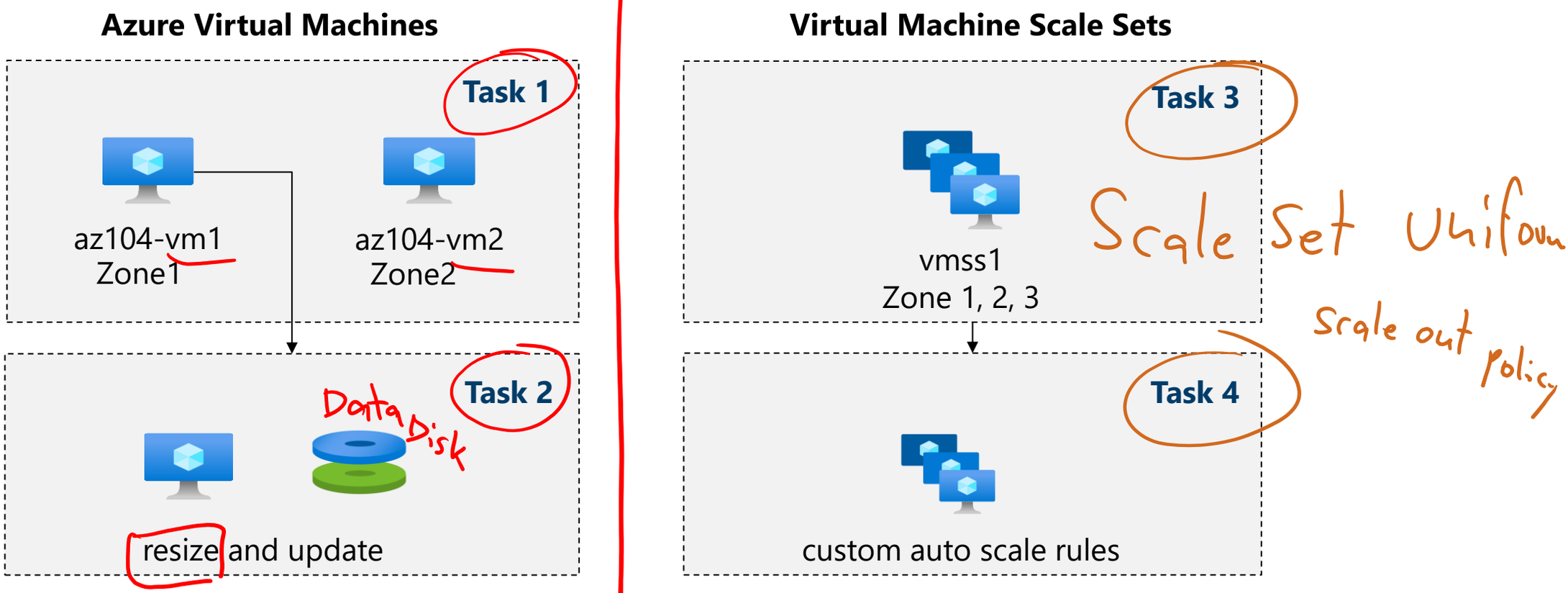
Lab 08 – Architecture diagram



Task 5: Create a virtual machine using Azure PowerShell (option 1)
Task 6: Create a virtual machine using the CLI (option 2)

New-AzVm ....
az vm create ....

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End of presentation

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